Diffusion models

Forward and backward process, generation, training, conditioning

Carlos Cotrini

CAS Building ML/AI Applications, ETH Zurich, weekend 4

Friday 23 October 2026

Which are real? Which are fake?



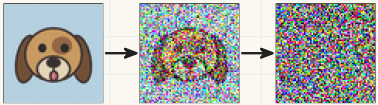
Diffusion models in brief

A diffusion model generates new data, such as images, by learning to undo a process that gradually adds noise.

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Training

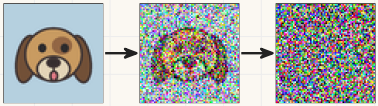


Take real images, add noise in many small steps, and learn to predict the noise that was added.

Diffusion models in brief

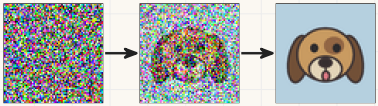
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Training



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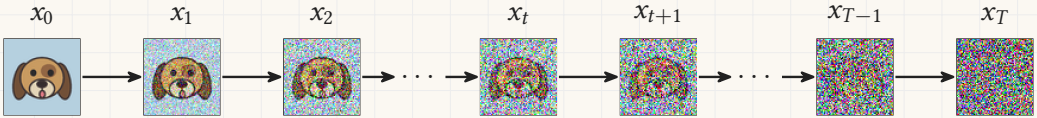
Generation



Start from pure noise and remove the predicted noise, step by step, until a clean image appears.

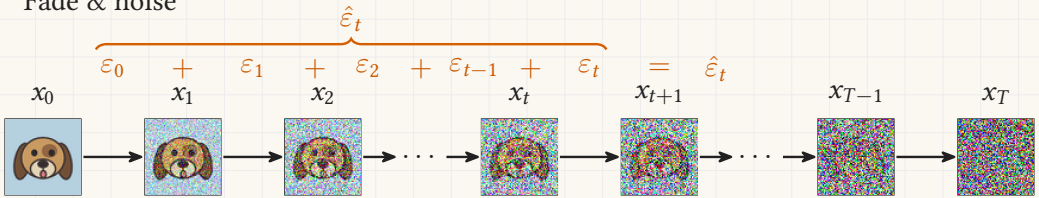
Forward & backward process

Fade & noise

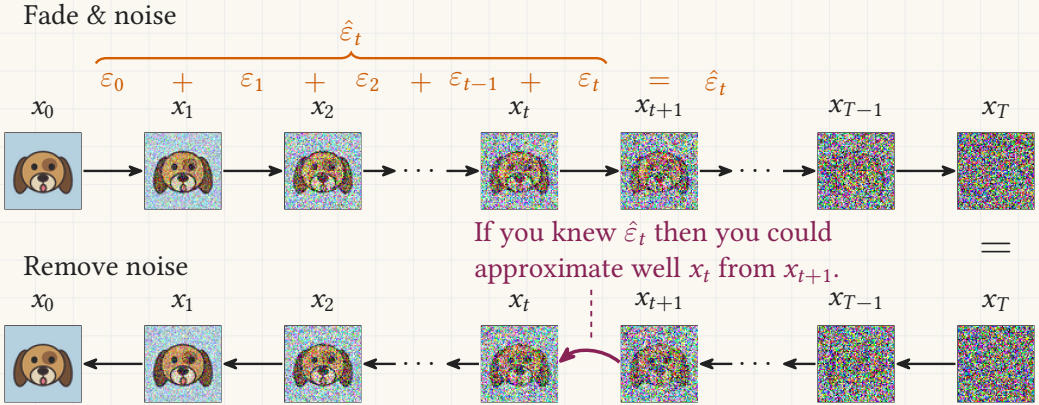


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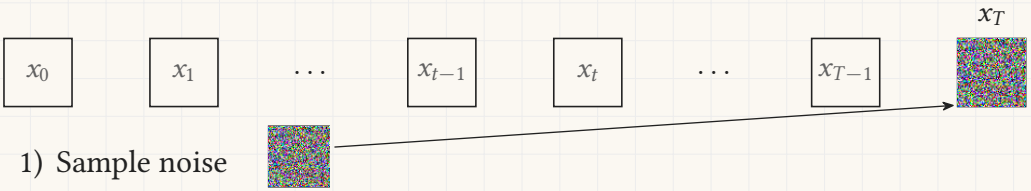
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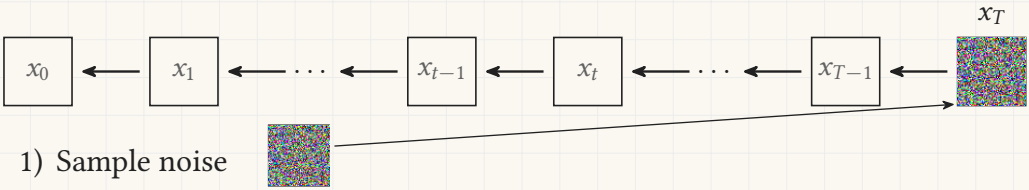
Forward & backward process



Generating from diffusion



Generating from diffusion

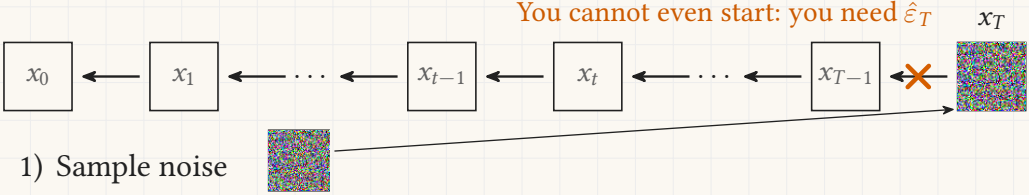


1) Sample noise

2) For $t = T, T-1, T-2, \ldots, 1$:

3) Output x_0

Generating from diffusion



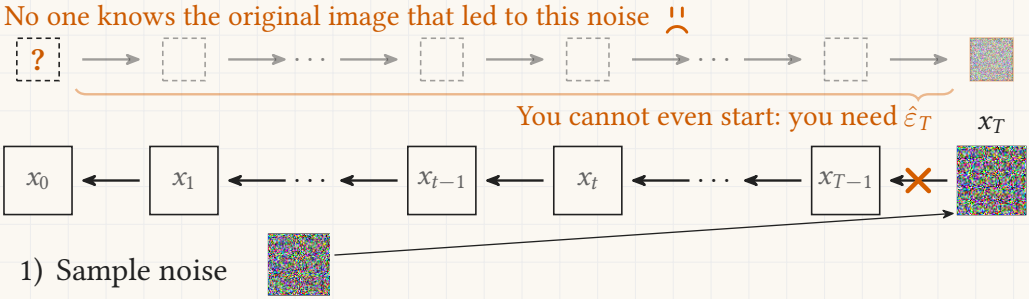
1) Sample noise

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   Estimate x_{t-1} from x_t using $\hat{\varepsilon}_t$

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Generating from diffusion



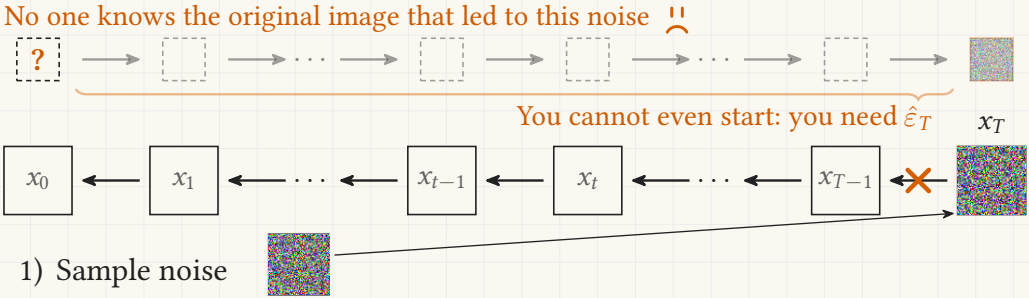
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   Estimate x_{t-1} from x_t using $\hat{\varepsilon}_t$ — You don't have it

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Training of a diffusion model

The model is trained to predict the noise $\hat{\varepsilon}_t$ from the noisy image x_t and the step t.

1. Take an image of the dataset.

x_0



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2. Pick $t \leq T$ at random.

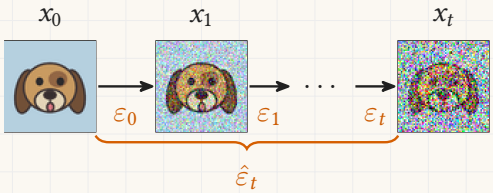
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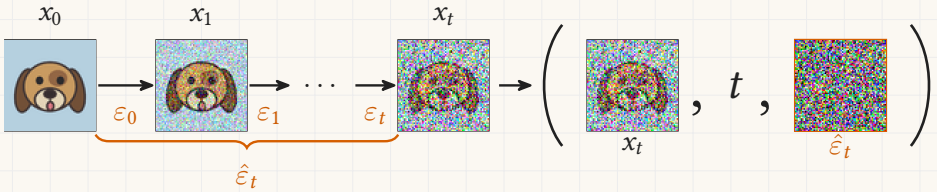
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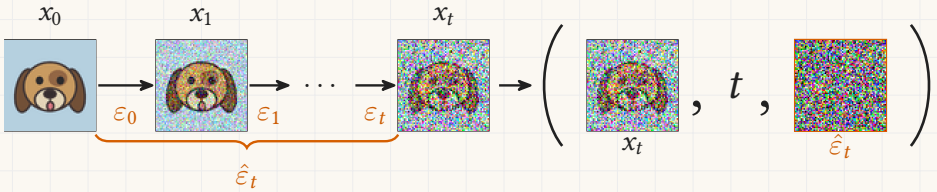
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Repeat until you collect $\{(x_t^{(1)}, \hat{\varepsilon}_t^{(1)}), \ldots, (x_t^{(n)}, \hat{\varepsilon}_t^{(n)})\}$

Training a diffusion model

$\hat{\varepsilon}_{t_1}$  ( , t_1)

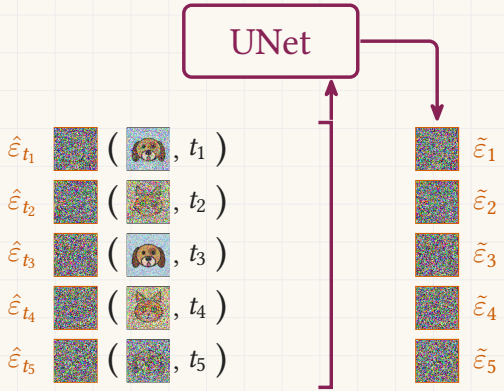
$\hat{\varepsilon}_{t_2}$  ( , t_2)

$\hat{\varepsilon}_{t_3}$  ( , t_3)

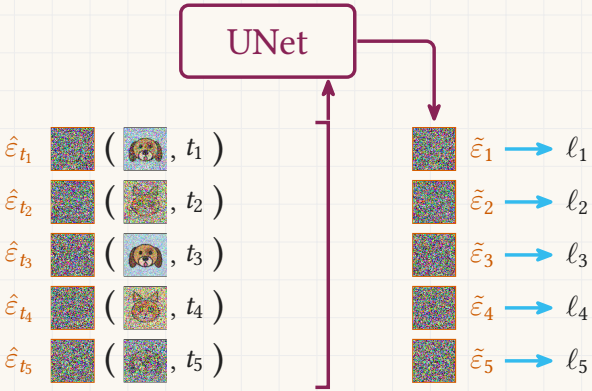
$\hat{\varepsilon}_{t_4}$  ( , t_4)

$\hat{\varepsilon}_{t_5}$  ( , t_5)

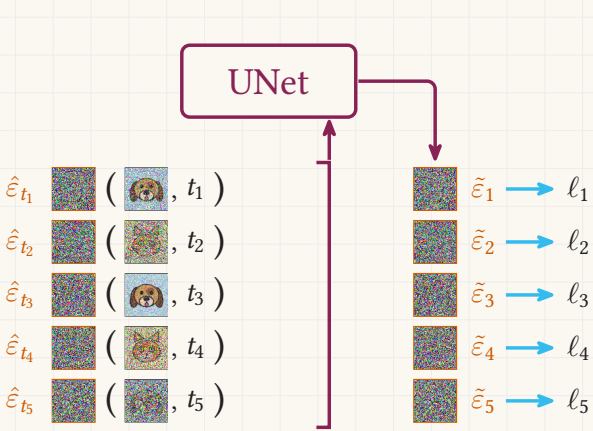
Training a diffusion model

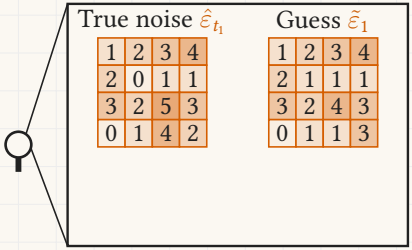


Training a diffusion model

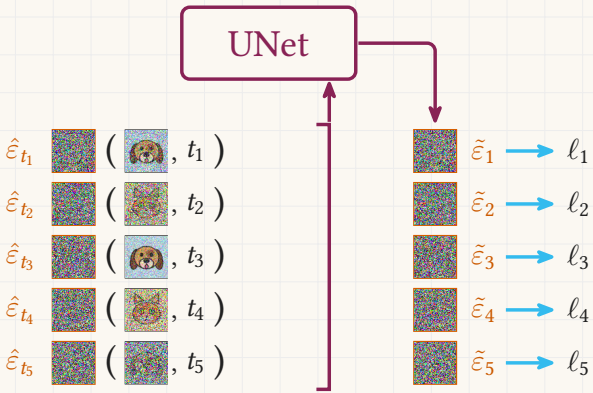


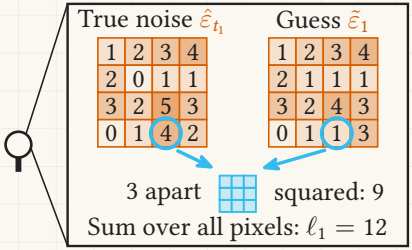
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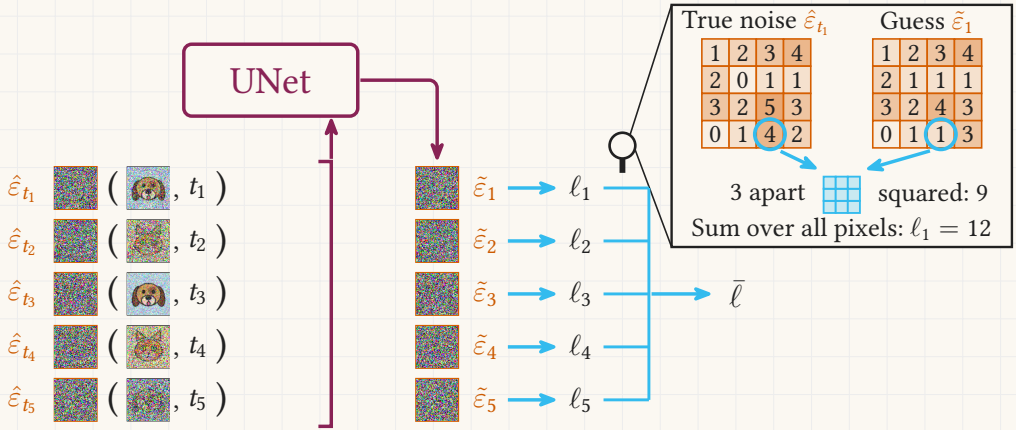


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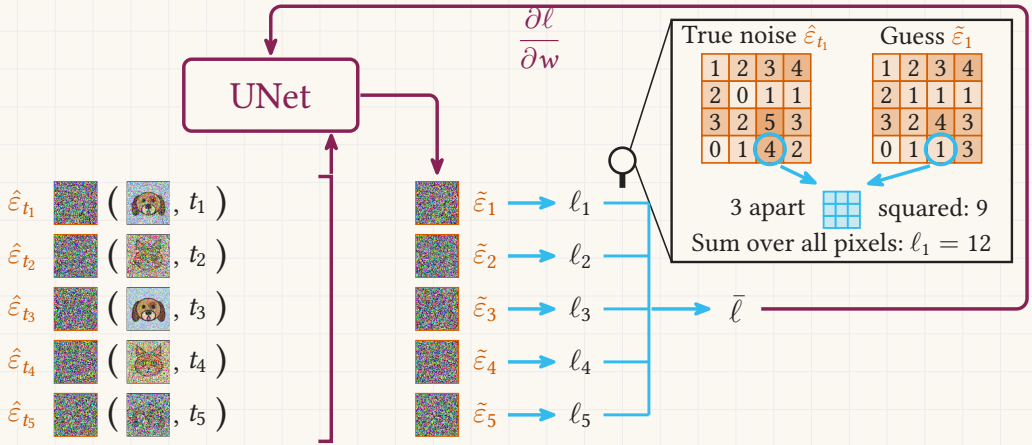




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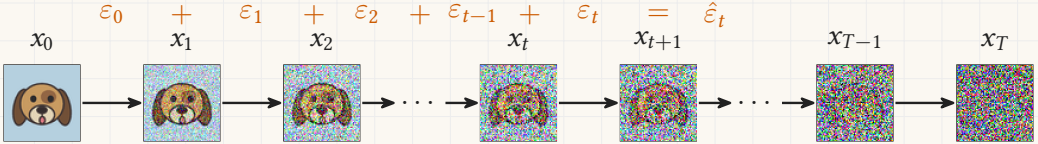
Training a diffusion model



Conditional diffusion models

The prompt "cat" is given to every backward step, so the image it produces is a cat.

Fade & noise

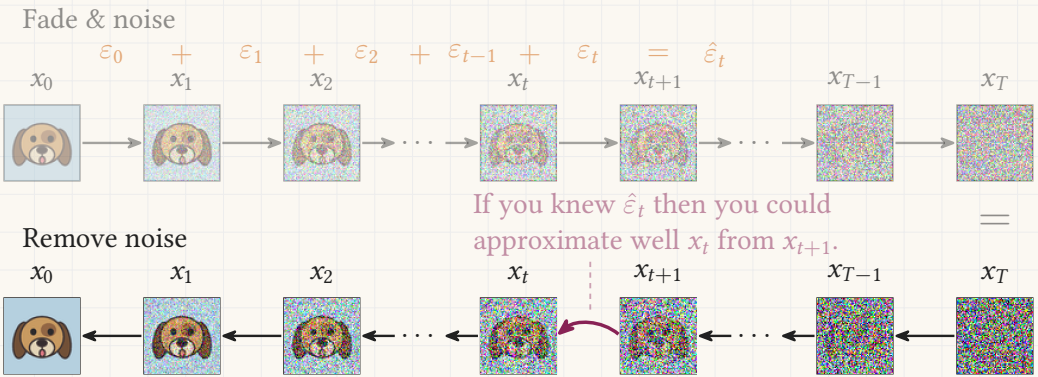


If you knew $\hat{\varepsilon}_t$ then you could approximate well x_t from x_{t+1}.

=

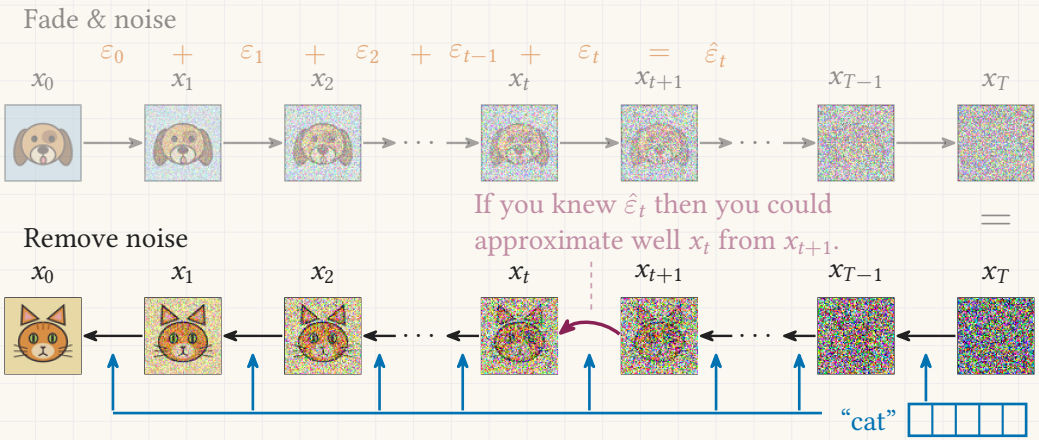
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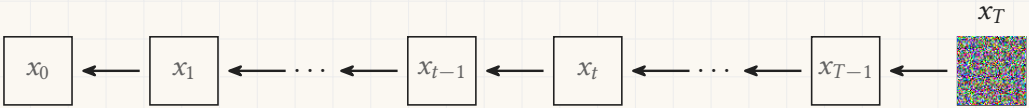
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Generating from diffusion

The prompt "cat" tells every backward step what the image should show.

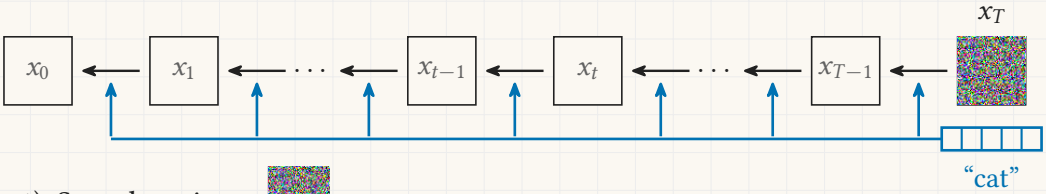


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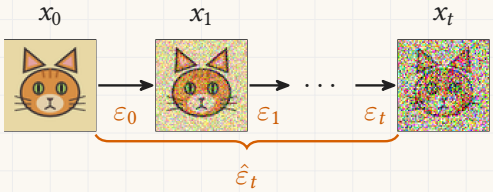
   Estimate x_{t-1} from x_t and "cat" using $\hat{\varepsilon}_t$ ← You don't have it ☹

3) Output x_0

Training of a diffusion model

Each training example also carries the caption of its image.

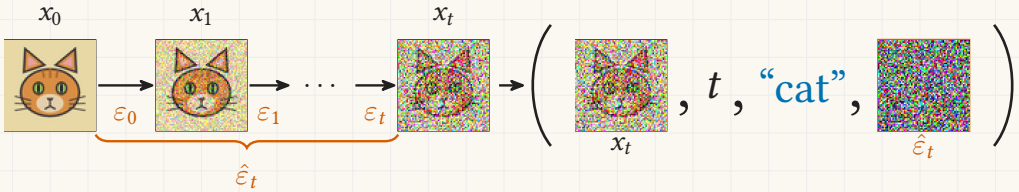
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Training of a diffusion model

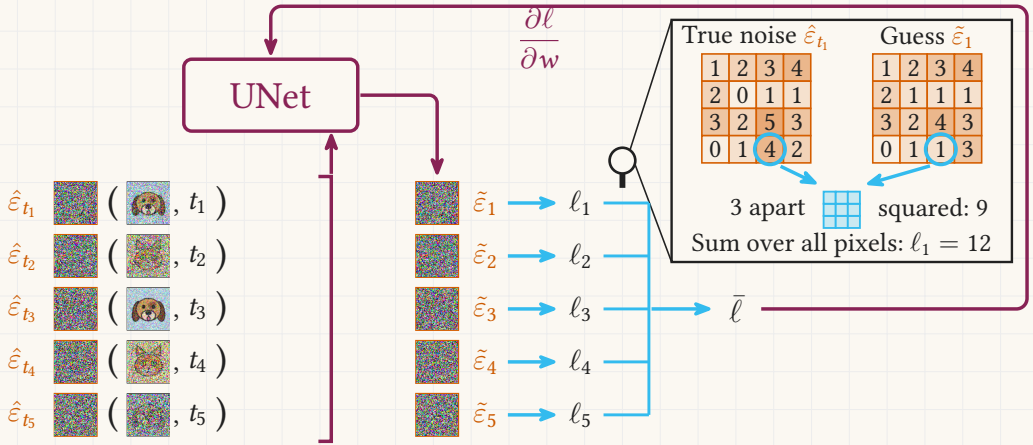
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